

A rank-one obstruction to random-sign sampling of the Grothendieck scale

Candidate solution packet for Question 3.7 of arXiv:2210.03868

13 August 2026

Status. Candidate full negative answer, likely valid, awaiting specialist review.

1 Source question

Sinclair and Vivek ask the following in Question 3.7 of [1], on page 10 of the source PDF.

Question 3.7. If $K = O(n)$ elements $\epsilon_1, \dots, \epsilon_K$ are sampled from $\{\pm 1\}^n$ independently does the Schur norm of the $K \times K$ matrix $[\langle A\epsilon_i, \epsilon_j \rangle]$ well approximate $\|A\|_{\infty \rightarrow 1}$ with a high degree of probability?

In symbols, if $\epsilon_1, \dots, \epsilon_K$ are independent and uniform in $\{\pm 1\}^n$, the question asks whether

$$B = (\langle A\epsilon_i, \epsilon_j \rangle)_{i,j=1}^K$$

has Schur norm $\gamma_2(B)$ comparable, with high probability and with $K = O(n)$, to

$$\|A\|_{\infty \rightarrow 1} = \sup_{\|x\|_{\infty} \leq 1} \|Ax\|_1.$$

Here

$$\gamma_2(C) = \inf_{C_{ij} = \langle x_i, y_j \rangle} (\max_i \|x_i\|_2) (\max_j \|y_j\|_2).$$

2 Proof intuition

Uniform signs see a fixed direction only at its typical fluctuation scale. Choose A to concentrate all of its $\ell_{\infty} \rightarrow \ell_1$ norm in the all-ones direction: $A = \mathbf{1}\mathbf{1}^T$. Its true norm is n^2 , attained by the single sign vector $\mathbf{1}$. A random sign vector has coordinate sum only about $\sqrt{n}$, however. The sampled matrix becomes the rank-one outer product of these coordinate sums, whose Schur norm is exactly the square of their maximum. Among only polynomially many samples this is $O(n \log n)$, a factor $O(\log n/n)$ below the true norm.

3 Two elementary lemmas

Lemma 1. For $u \in \mathbb{R}^r$ and $v \in \mathbb{R}^s$,

$$\gamma_2(uv^\top) = \|u\|_\infty \|v\|_\infty.$$

Proof. The one-dimensional factorization $(uv^\top)_{ij} = \langle u_i, v_j \rangle$ gives the upper bound. Conversely, every factorization $C_{ij} = \langle x_i, y_j \rangle$ satisfies

$$\max_{i,j} |C_{ij}| \leq (\max_i \|x_i\|_2)(\max_j \|y_j\|_2).$$

Taking the infimum and using $\max_{i,j} |u_i v_j| = \|u\|_\infty \|v\|_\infty$ proves the reverse bound. $\square$

Lemma 2. If $S = \sum_{r=1}^n \varepsilon_r$ for independent uniform signs, then for all $t \geq 0$,

$$\mathbb{P}\{|S| \geq t\} \leq 2 \exp\left(-\frac{t^2}{2n}\right).$$

Proof. For $\lambda > 0$,

$$\mathbb{E}e^{\lambda S} = (\cosh \lambda)^n \leq e^{n\lambda^2/2}.$$

Markov's inequality and the choice $\lambda = t/n$ give $\mathbb{P}\{S \geq t\} \leq e^{-t^2/(2n)}$. Apply the same argument to $-S$ and add the two bounds. $\square$

4 Counterexample and quantitative failure

Theorem 1. For each n , let $A_n = \mathbf{1}_n \mathbf{1}_n^\top$. Given independent uniform $\varepsilon_1, \dots, \varepsilon_K \in \{\pm 1\}$, set

$$B_{ij} = \langle A_n \varepsilon_i, \varepsilon_j \rangle.$$

Then, writing $S_i = \langle \mathbf{1}_n, \varepsilon_i \rangle$,

$$\|A_n\|_{\infty \rightarrow 1} = n^2, \quad \gamma_2(B) = \max_{1 \leq i \leq K} |S_i|^2.$$

Consequently, for every $0 < \delta < 1$,

$$\mathbb{P}\left\{\frac{\gamma_2(B)}{\|A_n\|_{\infty \rightarrow 1}} > \frac{2 \log(2K/\delta)}{n}\right\} \leq \delta, \quad (1)$$

and

$$\mathbb{E} \frac{\gamma_2(B)}{\|A_n\|_{\infty \rightarrow 1}} \leq \frac{2(\log(2K) + 1)}{n}. \quad (2)$$

In particular, if $K \leq Cn$ then the sampled Schur norm is $o(\|A_n\|_{\infty \rightarrow 1})$ with high probability. More generally, the ratio converges to zero in probability whenever $\log K = o(n)$.

Proof. For $\|x\|_\infty \leq 1$,

$$A_n x = \langle \mathbf{1}_n, x \rangle \mathbf{1}_n, \quad \|A_n x\|_1 = n |\langle \mathbf{1}_n, x \rangle| \leq n^2,$$

with equality at $x = \mathbf{1}_n$. Thus $\|A_n\|_{\infty \rightarrow 1} = n^2$.

Next,

$$B_{ij} = \langle \mathbf{1}_n \mathbf{1}_n^\top \varepsilon_i, \varepsilon_j \rangle = S_i S_j.$$

Hence $B = ss^\top$ for $s = (S_1, \dots, S_K)$, and Lemma 1 gives $\gamma_2(B) = \|s\|_\infty^2$.

By Lemma 2 and the union bound,

$$\mathbb{P}\{\gamma_2(B) \geq t^2\} \leq 2Ke^{-t^2/(2n)}.$$

Taking $t^2 = 2n \log(2K/\delta)$ proves (1). For the mean, put $Z = \gamma_2(B)$ and $u_0 = 2n \log(2K)$. Tail integration gives

$$\begin{aligned} \mathbb{E}Z &= \int_0^\infty \mathbb{P}\{Z \geq u\} du \\ &\leq u_0 + \int_{u_0}^\infty 2Ke^{-u/(2n)} du = 2n(\log(2K) + 1), \end{aligned}$$

which is (2) after division by n^2 .

Finally, for fixed $\eta > 0$ the same union bound directly yields

$$\mathbb{P}\left\{\frac{\gamma_2(B)}{n^2} \geq \eta\right\} \leq 2Ke^{-\eta n/2}.$$

This tends to zero whenever $\log K = o(n)$. □

5 Interpretation, scope, and verification

Question 3.7 uses the informal phrase “well approximate.” Theorem 1 rules out every customary dimension-free multiplicative interpretation: for the positive semidefinite rank-one family A_n , the sampled-to-true norm ratio itself tends to zero. It also shows that the obstruction persists for every subexponential number of samples.

The packet does not answer the independent noncommutative questions in Section 4 of the source paper. A companion verifier checks the exact matrix identity and reports seeded Monte Carlo ratios; it is a sanity check, not part of the proof.

A bounded novelty search through 13 August 2026 covered the run registry, local full-text indexes, the exact wording of Question 3.7, arXiv:2210.03868, the title and authors, and web combinations of “random signs,” “Schur norm,” and “Grothendieck norm.” No explicit answer or this rank-one obstruction was found. This is not an exhaustive priority search.

Human-review recommendation. Treat this as a likely valid full negative answer to Question 3.7. The key items to verify are Lemma 1, the normalization $\|A_n\|_{\infty \rightarrow 1} = n^2$, and that the source’s Schur norm is the same γ_2 used here.

References

- [1] T. Sinclair and N. Vivek, *Remarks on the Grothendieck norm*, arXiv:2210.03868 (2022).